

Developing Generative AI Solutions

Capabilities of Gen AI:

- Adaptability
- Responsiveness
- Simplicity
- Creativity and exploration
- Data efficiency
- Personalization
- Scalability

Challenges of Gen AI

- Regulatory violations
- Social risks
- Data security and privacy concerns
- Toxicity
- Hallucinations
- Interpretability
- Nondeterminism

Gen AI application Lifecycles

1. **Define a case:** analyze the application's functionalities, user needs, and business goals
2. **Select a foundation model:** select either existing pre-trained models or develop from scratch. Depend on the factors such as the availability of pre-trained models, the complexity of the case, and domain-specific data for training
3. **Improve performance:** fine-tuning. implement any necessary customizations or optimizations
4. **Evaluate the results:**
5. **Deploy the application**

Addressing Business Use Cases with Gen AI

- Cost savings
- Time savings
- Quality improvement
- Customer satisfaction
- Productivity gains: it can augment human capabilities, leading to increased productivity.

1. Define a case

Approaches:

- **Process automation:** to automate repetitive or time-consuming tasks, such as content generation and data analysis, or custom-service interaction
- **Augmented decision-making:** providing insights, recommendations, and decision support
- **Personalization and customization:** customized content, products. experience for the customer
- **Creative content generation:** text, video, audio
- **Exploratory analysis and innovation:** explore new data, concepts, ideas, solutions

2. Selecting an FM (Foundation model)

Some criteria to consider include the following:

- **Cost:** it can be expensive, including licensing fees, computational powers, potential customization, and fine-tuning costs
- **Modality:** models can be designed for different modalities, such as text generation, image generation, audio generation, or multimodal generation. It depends on the output format and the target application
- **Latency:** Some applications require real-time or low-latency generation
- **Multi-lingual support:** requires generating content in multiple languages
- **Model size:** scale
- **Model Complexity:** Based on the transformer architectures or LLMs, they can handle more advanced tasks, but might be more challenging to deploy and optimize
- **Customization:** ability to fine-tune, adapt to specific domains or tasks
- **Input/Output length:** Generative models might have limitations on the maximum input or output sequence lengths that they can handle.
- **Responsibility considerations:** It's important to evaluate the responsible implications of using pre-trained generative AI models, such as potential biases, misinformation risks, or misuse. Models should be vetted for their training data sources and potential societal impacts.
- **Deployment and integration:** The ease of deployment, compatibility with existing infrastructure, and availability of tools or libraries for integrating the model into the target application should be considered.

3. Improving the performance of an FM

- **Prompt engineering:** System prompt, user prompt, to adjust the output behavior based on the specific context of the output that you want to achieve. Prompt engineering refers to the process of carefully crafting the input prompts or instructions given to the model to generate desired outputs or behaviors.

- **Design:** Crafting clear, unambiguous, and context-rich prompts that effectively communicate the desired task or output to the model
- **Augmentation:** Incorporating additional information or constraints into the prompts, such as examples, demonstrations, or task-specific instructions, to guide the model's generation process
- **Tuning:** Iteratively refining and adjusting the prompts based on the model's outputs and performance, often through human evaluation or automated metrics
- **Ensembling:** Combining multiple prompts or generation strategies to improve the overall quality and robustness of the outputs
- **Mining:** Exploring and identifying effective prompts through techniques like prompt searching, prompt generation, or prompt retrieval from large prompt libraries

Prompt technique:

- Zero-shot prompting
- Few-shot prompting
- Chain-of-thought (CoT) prompting
- Self-consistency
- Tree of Thoughts (ToT)
- Retrieval Augmented Generation (RAG)
- Automatic Reasoning and Tool-use (ART)
- ReAct prompting

- RAG:

- It is an NLP technique that combines the capabilities of retrieval systems and generative language models to produce high-quality and informative text outputs.

- It has two main components:

- **Retrieval system:** This component retrieves relevant information from a large corpus of text data, such as a knowledge base, web pages, or other textual sources. The retrieval system uses techniques like information retrieval, sparse indexing, or dense retrieval to identify the most relevant passages or documents for a given input query or context.

- **Generative language model:** This component is a large pre-trained language model, such as GPT-3, BART, or T5, that can generate natural language text. The language model takes the input query or context, along with the retrieved relevant information. And from this, it generates a coherent and fluent text output that combines the retrieved knowledge with its own understanding and language generation capabilities.

RAG Business Applications

- **Building intelligent question-answering systems**
- **Expanding and enriching existing knowledge bases:** generating new knowledge or rephrasing existing information in a more natural and understandable

way. This can improve the accessibility and usability of knowledge bases for various applications.

- **Generating high-quality content:** articles, reports, or summaries, by combining retrieved information from various sources with the language generation capabilities of the model. This can be useful in domains like journalism, research, or content marketing.

Fine-tuning

Fine-tuning is another way to improve the performance of a foundation model even further.

Fine-tuning refers to the process of taking a pre-trained language model and further training it on a specific task or domain-specific dataset. Fine-tuning allows the model to adapt its knowledge and capabilities to better suit the requirements of the business use case. Although FMs are pre-trained through self-supervised learning and have an inherent capability of understanding information, fine-tuning the FM base model can improve performance.

During fine-tuning, the model's parameters are updated to better capture the patterns and nuances present in the task-specific data.

Ways to fine-tune a model:

- Instruction fine-tuning uses examples of how the model should respond to a specific instruction. Prompt tuning is a type of instruction fine-tuning.
- Reinforcement learning from human feedback (RLHF) provides human feedback data, resulting in a model that is better aligned with human preferences.

4. Evaluation

- **Human evaluation:** This can involve tasks such as open-ended conversations, question-answering, text generation, or other specific use cases. Human evaluators can provide qualitative feedback on factors like coherence, relevance, factuality, and overall quality of the model's outputs. Although human evaluation is often considered the gold standard, it can be time-consuming and expensive, especially for large-scale evaluations.
- **Benchmark datasets:** curated collections of data designed specifically for evaluating the performance of language models or other AI systems. These datasets often consist of carefully selected examples or tasks that cover a wide range of topics, complexities, and linguistic phenomena. Models are evaluated by running them on these benchmark datasets and measuring their performance using predefined metrics or tasks.
 - Popular benchmark datasets for NLP tasks include the following:
 - **The General Language Understanding Evaluation (GLUE)** benchmark is a collection of datasets for evaluating language understanding tasks like text classification, question answering, and natural language inference.

- **SuperGLUE** is an extension of GLUE with more challenging tasks and a focus on compositional language understanding.
- **Stanford Question Answering Dataset (SQuAD)** is a dataset for evaluating question-answering capabilities.
- **Workshop on Machine Translation (WMT)** is a series of datasets and tasks for evaluating machine translation systems.
- **Automated metrics:** Automated metrics can be useful for rapid iterations and fine-tuning during model development, but they often fail to capture the nuances and complexities of human language and might not align perfectly with human judgments.
 - **Perplexity** (a measure of how well the model predicts the next token)
 - **BLEU score** (for evaluating machine translation)
 - **F1 score** (for evaluating classification or entity recognition tasks)
 - **BERTScore** is a metric that evaluates the semantic similarity between a generated text and one or more reference texts. It uses pre-trained Bidirectional Encoder Representations from Transformers (BERT) models to compute contextualized embeddings for the input texts, and then calculates the cosine similarity between them.
 - **Recall-Oriented Understudy for Gisting Evaluation (ROUGE)** is a set of metrics used for evaluating automatic summarization and machine translation systems. It measures the quality of a generated summary or translation by comparing it to one or more reference summaries or translations.

5. Deploy the model

Key consideration:

- **Cost:** Pay for the resources that you use with no minimum fees.
- **Regions:** Model deployment is limited to certain AWS Regions.
- **Quotas:** Ensure that you have adequate service resources for your AWS account.
- **Security:**
 - If your model is deployed in AWS infrastructure, the security responsibility is shared between the company and AWS.
 - If accessing a model outside of AWS, security considerations must be evaluated for data leaving the AWS account.